

A fixed-character obstruction to PAH-v2 fine-Gibbs generator convergence

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope: exact question and frozen source

PAH-V2-GD-002 asks, for every fixed admissible parameter tuple, fixed h, N , base r_0 and full invariant finite observable f_0 , whether

$$\lim_{R \rightarrow \infty} \sup_{s > r \geq R} \|L_s I_{r,s} f_r - I_{r,s} L_r f_r\|_{L^2(\pi_s)} = 0, \quad f_r = I_{r_0,r} f_0. \quad (1)$$

The weight is the original full counting Gibbs state $\pi_s(x) = e^{-\beta F_s(x)} / \sum_y e^{-\beta F_s(y)}$. Neither a selected observable subalgebra nor a different state is used. The inherited sup-norm algebra remains unchanged; (1) is a separately authorized diagnostic, not a declared limiting-generator domain. The preregistration was written and hash-checked before execution:

strategy/pa-hyp/PAH-v2-GD-002-prereg-v1.json
810ac2d0f787bfdc41a4254c148e327d682cc4eaa0fb4112feaf87064757ce9a

The approved comparison bundle is still

1d6fa47f5351d74f05bed3aba5b1e229a827644d4c6ef5d7dd73c117adee95b6

The source locators are PAH-001-v1, `functional_or_action` and `dynamics`; PAH-001-v2-r2, `root_partial_maps` and `operators_and_pairings`; and the adopted Morton/CRT draft, `regulators`, `state_maps`, `observable_maps`. All exact hashes are in the result certificate. R-570 and R-571 supply only their finite operator and comparison definitions. R-572 remains a distinct sup-negative result; its Gibbs- L^2 field was not evaluated.

Fix the existing $h = N = 0$ open square, with four vertices, four stored edges, one bounded face, and label-only anchors $O = (0, 0)$, $C = (1, 0)$. Fix $r_0 = 2$, $q_0 = m_0 = R_0 = 1$, and

$$\epsilon = \beta = \nu = \lambda_s = \eta_6 = \kappa_s = \kappa_D = \kappa_g = g = 1, \quad m^2 = \lambda_4 = 0. \quad (2)$$

Every rate scale is unity as an inserted fixed input, not a fitted output. The approved schedule is $K_r = \prod_{i=1}^{r+1} p_i$, $M_{s,r} = 2^r$, $M_{\psi,r} = 4^r$, $R_{\max,r} = 2^r$, $Q_r = 2^r$. All integer coordinates, including hidden aperture and zero-amplitude phase labels, remain distinct. The time is unchanged external stochastic Markov time, with the original stationary initial-value boundary. Only cutoff r is varied; later lattice h , exhaustion N , and inherited limits are not taken.

2. One fixed admitted character

Let $\Phi(x) = \sum_{e \in \partial p} \sigma_e u_e$ be the signed integer link sum around the sole face, and put

$$f_0(x) = \exp(2\pi i \Phi(x)/5), \quad a_r = (K_r/K_2)^{-1} \pmod{5}, \quad f_r(x) = \exp(2\pi i a_r \Phi(x)/5). \quad (3)$$

Since $K_2 = 30$ and $5 \mid K_r$, these are well-defined functions of the original modular link coordinates. The signed sum of each gauge coboundary vanishes around the closed face; wrap corrections are multiples of K_r , also invisible modulo five. Thus the functions are gauge invariant. The full anchor-preserving cell group of this anchored square is trivial: fixing the adjacent labelled vertices fixes their two other neighbours. No symmetry subgroup or physical projection is substituted.

The admitted cutoff projection multiplies links by P^{-1} modulo K_r , where $P = p_{r+2}$. Its reduction modulo five gives exactly $a_{r+1} = a_r P^{-1} \pmod{5}$ and (3). Composition gives every nonadjacent pullback. Thus f_0 , not a cutoff-dependent selection, is fixed once and for all, and $|f_r| = 1$ on every full counting state.

PH, TR and AP do not change any link coordinate, so their observable increments in (3) vanish individually. Their rates are not set to zero and their incidences are not removed. Each of the eight signed LK incidences is valid, including modular wrap. A unit-rate algebraic leading part therefore is

$$\lambda(a) = 4\{2\cos(2\pi a/5) - 2\}, \quad L_r f_r = \lambda(a_r) f_r + e_r. \quad (4)$$

Equation (4) is a decomposition of the original generator, not a replacement by a unit-rate model. The remainder e_r is bounded next using the actual rates.

3. Uniform original-rate remainder on all states

Write $A_v = 2^{-r} \ell_v$. Then $A_v \geq 0$ and $\sum_v A_v = 1$ on every source state. At (2), every aperture is one, every J_e, J_p is one, and all mobilities are one. Under one LK incidence the only changing functional terms are its matter edge and the one plaquette containing it. With $\delta = 2\pi/K_r$ and $|e^{i\delta} - 1| \leq |\delta|$, the matter change has absolute value at most $A_v A_w \delta \leq \delta$, and the plaquette change has absolute value at most δ . Onsite and all other terms are unchanged. Hence, uniformly in every occupation, phase, aperture and link label,

$$|F_r(bx) - F_r(x)| \leq 4\pi/K_r \leq 16/K_r. \quad (5)$$

The original rate is $c_b = e^{-\Delta F_b/2}$. For $K_r \geq 8$, set $u = 8/K_r \leq 1$. Convexity gives $e^u - 1 \leq (e - 1)u \leq 2u$ on $[0, 1]$; thus

$$|c_b - 1| \leq 16/K_r, \quad |e_r(x)| \leq 8 \cdot 2 \cdot 16/K_r = 256/K_r. \quad (6)$$

The factor eight counts both signs on all four source edges; the factor two bounds a unit character difference. There is no generator half factor. The half in B belongs only to the Dirichlet form convention. For $s = r + 1$, exact comparison of (4) gives

$$\Delta_{r,s} f_r = (\lambda(a_s) - \lambda(a_r)) f_s + e_s - e_r \circ p_{s,r}, \quad |e_s - e_r \circ p_{s,r}| \leq 512/K_r. \quad (7)$$

No Gibbs projectivity or norm comparison between different states was used.

4. Arbitrarily late CRT flips and the Gibbs lower bound

The nonzero residues modulo five split into $\{1, 4\}$ and $\{2, 3\}$. Multiplication by the inverse of a residue in $\{2, 3\}$ swaps these two sets. The elementary fifth-root cosine values give

$$\lambda(a) \in \{-10 + 2\sqrt{5}, -10 - 2\sqrt{5}\}.$$

Whenever $p_{r+2} \equiv 2$ or $3 \pmod{5}$,

$$|\lambda(a_{r+1}) - \lambda(a_r)| = 4\sqrt{5} > 8. \quad (8)$$

An elementary Euclid construction supplies such primes beyond every bound; no asymptotic prime-distribution theorem is imported. Given any prime bound $B \geq 11$, put $P_B = \prod_{p \leq B, p \neq 5} p$ and choose

$t \in \{1, 2, 3, 4\}$ with $tP_B \equiv 1 \pmod{5}$. The integer $n_* = 1 + tP_B$ is 2 modulo five and 1 modulo every prime at most B other than five. All its prime factors exceed B . If each factor were 1 or 4 modulo five, so would their product be, a contradiction. Consequently some prime factor is 2 or 3 modulo five.

For any tail threshold R , choose $B = p_{\max(R, 4)+1}$ and such a prime $q = p_k > B$. Put $r = k - 2$ and $s = r + 1$. Then $r \geq \max(R, 4)$ and the flip condition holds. Moreover $K_r \geq K_4 = 2310 > 512$. Equations (7)–(8) imply, for *every* original fine state x ,

$$|\Delta_{r,r+1}f_r(x)| \geq 8 - 512/K_r \geq 7. \quad (9)$$

The actual Gibbs weights are positive and sum to one, so

$$\sum_x \pi_{r+1}(x) |\Delta_{r,r+1}f_r(x)|^2 \geq 49. \quad (10)$$

This proves the exact fixed-data negation of (1), with $\eta_0 = 7$ and arbitrarily late adjacent pairs. Adjacent pairs are used as witnesses to a universal all-pair claim, not as sufficient evidence of Cauchy convergence. In particular this obstruction does not live only on a Gibbs-negligible set. All constants in (5)–(10) are derived conservative bounds, not fitted values.

5. State consistency and nontriviality, separately

State projectivity, a common limiting pairing and semigroup convergence are not proved or disproved by (10); the argument does not require them. For nontriviality only, let γ_r denote normalized uniform counting on the same original finite set. This auxiliary measure is used in a domination inequality, never substituted for π_r in (1) or (10). The displayed functional at (2) is nonnegative and satisfies

$$0 \leq F_r \leq \frac{1}{6} + \frac{1}{2} + \frac{4}{2} + 2 = \frac{14}{3}. \quad (11)$$

Indeed $\sum A_v^6 \leq 1$, $\sum A_v^2 \leq 1$, each matter edge contributes at most $(A_v + A_w)^2/2 \leq 1/2$, and the plaquette contributes at most two. Thus $\pi_r(x) \geq e^{-14/3} \gamma_r(x)$ uniformly. Independent uniform link counting makes $\Phi \bmod 5$ uniform, so $\mathbb{E}_{\gamma_r} f_r = 0$ and

$$\text{Var}_{\pi_r}(f_r) = \inf_{z \in \mathbb{C}} \mathbb{E}_{\pi_r} |f_r - z|^2 \geq e^{-14/3} > 0. \quad (12)$$

For $K_r \geq 8$, (5) also gives $c_b \geq e^{-1}$ on every LK incidence. With the original root half convention,

$$\mathcal{E}_r(f_r, f_r) = \frac{1}{2} \sum_{x,b} \pi_r(x) c_b(x) |f_r(bx) - f_r(x)|^2 \geq 4e^{-1} > 0, \quad (13)$$

since $2(1 - \cos(2\pi a_r/5)) > 1$ and there are eight LK incidences. The fixed character therefore does not collapse to a constant or a null-energy observable along this sequence. This is not a construction of a limiting physical state or nontrivial limiting semigroup.

6. Independent formulation and Devil’s-advocate review

The independent implementation reconstructs the square, full anchored automorphism group, integer tuples and all signed partial maps without importing the primary evaluator or source backends. It evaluates the original finite functional and rates with directed Arb bounds, including nonzero links, nonzero phases, split occupation and aperture endpoints. An alternative analytic derivation diagonalizes the two translation operators on the five-residue character: each edge contributes $z^a + z^{-a} - 2$. The all-state triangle bound (6) then proves (7) without numerical rate fitting. The primary implementation uses the original source enumerator and exact cyclotomic polynomial reduction; neither finite fixture proves an infinite tail.

1. **Gauge or anchor failure? DISMISSED.** Closed-face telescoping, modular wrap divisibility and the full trivial anchored cell group establish admission of the fixed observable. It is not an open-link character.
2. **Dropped roots or replaced rates? DISMISSED.** PH/TR/AP have individual zero increments. Both LK signs remain. Equations (5)–(7) bound the original exponential rates; unit rates are a leading decomposition only.
3. **Small Gibbs mass? DISMISSED.** Equation (9) holds at every fine state. Equation (10) uses only the original normalized Gibbs weights.
4. **Finite or moving-observable counterexample? DISMISSED.** Equation (3) is a single fixed-base pullback; the Euclid construction gives pairs beyond each threshold, not a finite table or a new observable per pair.
5. **Time/factor mismatch? DISMISSED.** No time rescaling, volume normalization or generator half factor is used. The Dirichlet half is retained.
6. **State or dynamics overclaim? UPHELD as a boundary.** Projectivity, a limiting generator domain, semigroup convergence and all alternative topologies remain unsettled. R-570/R-571 and physical claims are not refuted.
7. **External/Lean overstatement? UPHELD as a boundary.** The two implementations and hostile review have same-task authorship. Five Lean declarations check parameterized residue/gap/perturbation/weight/arithmetic bridges; the concrete PAH analytic and prime-existence bridges are not fully encoded. External independent review is invited.

The literature-first bounded search considered Kuwae–Shioya, *Communications in Analysis and Geometry* 11 (2003), 599–673, doi:10.4310/cag.2003.v11.n4.a1, as background on varying Hilbert spaces. Applicability to a PAH limiting realization is NOT-YET-ASSESSED and no theorem is imported. The present proof instead derives the finite source formulas, uniform remainder and elementary residue/prime argument directly. There is no novelty claim and no Q3LOCK/TECT-YM evidence import.

7. Result footer and reproducibility

Result ID: R-573; exploration EXP-001741.

Precise statement: The fixed character (3) has fine-Gibbs defect norm at least seven at adjacent pairs beyond every cutoff threshold; GD-002 is DISPROVED.

Scope: Exact PAH-v2 and approved square/CRT comparison at (2); full counting states; original Markov time; cutoff-first stage only.

Dependencies: Hash-pinned PAH-v1/v2-r2 and admitted comparison R-571; finite operator conventions R-570; no imported limit theorem.

Evidence grade: ANALYTIC, EXACT, EXECUTED; partial parameterized Lean.

Reproduction command: One line assembled from the displayed command below.

Expected output: Primary, independent, hostile and partial Lean replay PASS; the mathematical target is DISPROVED, not passed by those checks.

Falsification gate: An invalid fixed character, omitted root, failed uniform rate bound, invalid CRT coefficient or prime-tail construction refutes this negative proof and requires an append-only correction.

Tier before / after: C6 remains T1; T-054's active gate is unchanged.

No-overclaim statement: No universal all-domain dynamics no-go, new physical projection, common Hilbert state, semigroup, lattice/volume or continuum result; no physical Pre-A, Sector-A, spacetime, QFT, gravity, Yang–Mills, mass gap or TOE.

Next required action: Does the same fixed character obstruct original-time semigroup comparison, or only generator-value comparison? This is a distinct future question requiring a separate target; it is not evaluated here.

```
python -X utf8 verification/scripts/pah_v2_gd002_verify.py
--lean-cache E:/Dev/TECT/verification/lean/.lake/packages
--elan-home C:/Users/NaEun/.elan
```

Reenter GD-002 only on an exact source/definition/inequality/tail error, not by repeating larger finite tables. One synthesis note is issued for this checkpoint; no intermediate-lemma PDF is required.